N-of-1 Longitudinal Study of a Patient with ME/CFS: Methods and Preliminary Results

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Abstract

Keywords: n-of-1; intensive longitudinal data; ME/CFS; fNIRS; EEG; HRV; POTS

1 Introduction

2 Methods

2.1 Study design

2.2 Data acquisition

2.2.1 Symptom scores

2.2.2 Physical activity

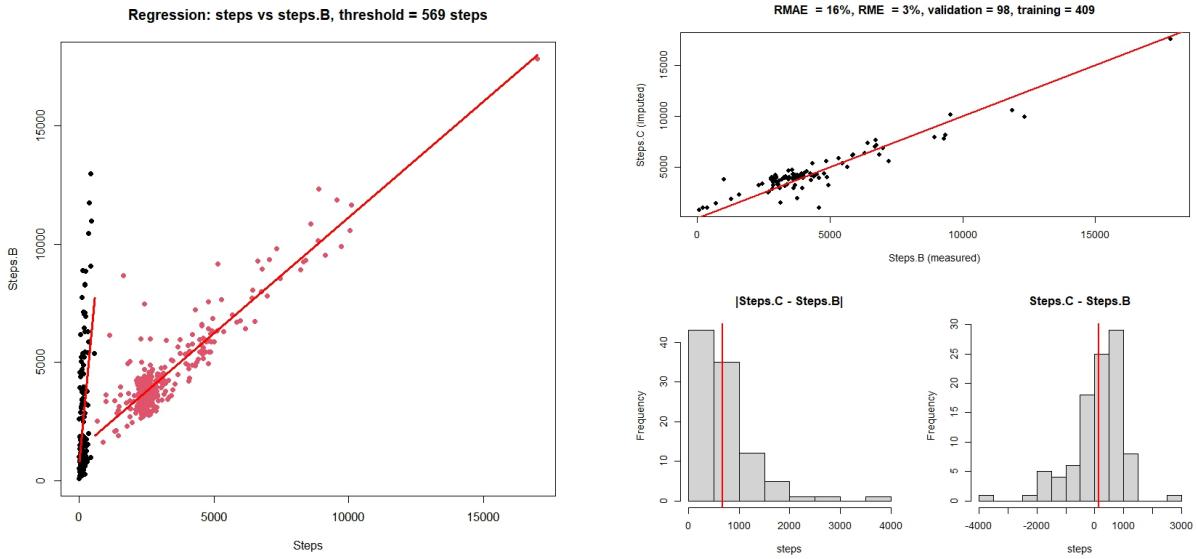
Daily step counts were recorded with a smartphone (Redmi Note 11S/Pedometer Step Counter) from January 2023, and with a wrist-worn watch (Dyurawa) from March 2025. The phone records only the steps taken while it is carried, so it misses most activity at home, whereas the watch records activity throughout the day; watch counts were therefore taken as the reference measure.

Gaps of up to four consecutive days were filled by linear interpolation in both series, if missing days were less than 10%. Days on which the watch recorded fewer steps than the phone were considered days of watch non-wear or malfunction, and their watch counts were set to missing.

To obtain a continuous series of watch-equivalent steps over the whole study period (`Steps.C`), watch counts were estimated from phone counts on days without watch data. The relationship between the two counts differs between days spent at home, when the phone is mostly not carried and its count carries little information, and days with outdoor activity, when the phone is carried. Days were therefore divided into two regimes by a threshold t on phone steps (`Steps`), and a linear regression of watch steps (`Steps.B`) on phone steps was fitted separately in each regime. The threshold was chosen by scanning values from 50 to 2000 phone steps and selecting the one that minimised the total sum of squared residuals of the two regressions, requiring at least 20

days in each regime (see Figure 1a). On days with watch data, `Steps.C` equals the measured watch count; on the other days, it is the value predicted by the regression of the corresponding regime.

The imputation was validated by fitting the model on the first 80% of the watch period and comparing the values predicted for the remaining 20% with the measured ones (Figure 1b). The relative mean absolute error, $\sum |\hat{y}_i - y_i| / \sum y_i$, and the relative mean error (bias), $\sum(\hat{y}_i - y_i) / \sum y_i$, are indicated on the plot. Most days were slightly overestimated, while the most active days, and outdoor activity without the phone, were underestimated. The time series of `Steps`, `Steps.B`, and `Steps.C` are shown in Figure 2.



(a) Regression model. (b) Validation.

Figure 1: Imputation of watch steps (`Steps.B`) from phone steps (`Steps`).

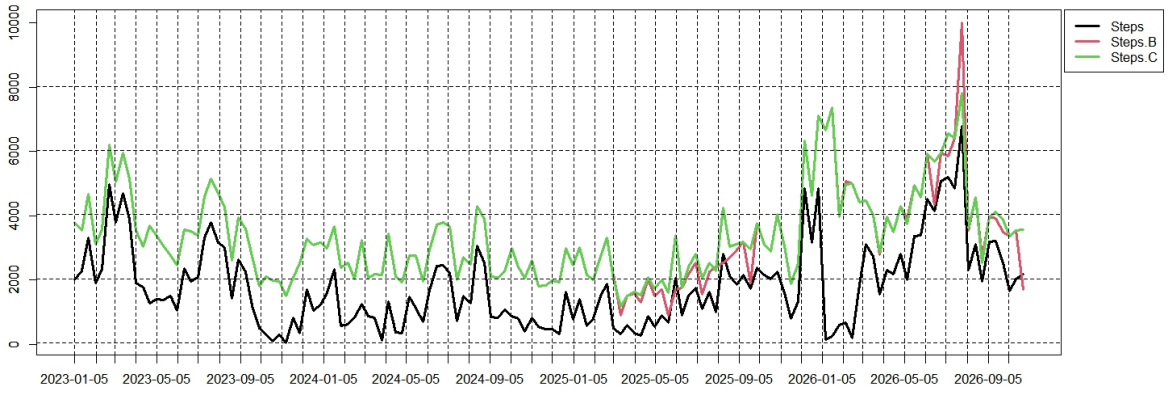


Figure 2: Daily steps averaged over non-overlapping 10-day blocks.

2.2.3 Heart rate variability

Heart rate variability (HRV) was assessed from five-minute recordings of RR intervals, following the protocol and the analysis pipeline described in detail in a previous work [1], to which the reader is referred for an overview on HRV, the definition of the indices, their physiological interpretation, and an introduction to spectral analysis by Discrete Fourier Transform. Recordings started in July 2022 and were performed once a day, in the morning, in the supine position, after five minutes of rest in the same position. RR intervals were acquired for about five minutes with a Polar H9 sensor mounted on a moistened chest belt, a type of device whose agreement with the electrocardiogram has been documented [2]. The sensor was connected via Bluetooth to a smartphone running the Elite HRV application, and the RR series were exported in text format, one file per recording.

The series were analysed with the R package RHRV [3]. For each recording, the instantaneous non-interpolated heart rate was computed from the RR intervals, and anomalous beats (ectopic beats and artefacts) were removed with the adaptive filter of the package, which discards beats that differ too much from the preceding ones or that fall outside physiological limits (default parameters). The filtered series was then linearly interpolated at 4 Hz to obtain the evenly sampled signal required by the discrete Fourier transform. Recordings shorter than 240 s or longer than 360 s were excluded, and when more than one recording was available for the same day only the first one was retained.

In the time domain, the following indices were computed: mean heart rate (HR) and mean RR interval (RR); the standard deviation of the RR intervals (SDNN) and the HRV triangular index (HRVi), as measures of overall variability; the root mean square of successive differences (rMSSD) and the percentage of successive differences greater than 50 ms (pNN50), as measures of the high-frequency, vagally mediated component [3, 4].

In the frequency domain, the power spectrum was estimated with the short-time Fourier transform, using a single window of 300 s, since for five-minute recordings shorter windows give less reliable estimates of the low-frequency power [2]. The power was integrated over the low-frequency (LF, 0.04 Hz to 0.15 Hz) and the high-frequency (HF, 0.15 Hz to 0.40 Hz) bands [5]. The ultra-low- and very-low-frequency bands were not considered, since they cannot be reliably estimated from five-minute recordings [5]. From LF and HF, the following derived quantities were computed: the ratio LF/HF; the total power, defined here as $\mathrm{TotPower} = \mathrm{LF} + \mathrm{HF}$; and the normalised powers $\mathrm{LF_{nu}} = \mathrm{LF}/(\mathrm{LF} + \mathrm{HF})$ and $\mathrm{HF_{nu}} = \mathrm{HF}/(\mathrm{LF} + \mathrm{HF})$, which reduce the within- and between-subject variability of the absolute powers [6]. The distributions of the time-domain and frequency-domain indices over all the recordings are shown in Figures 3 and 4.

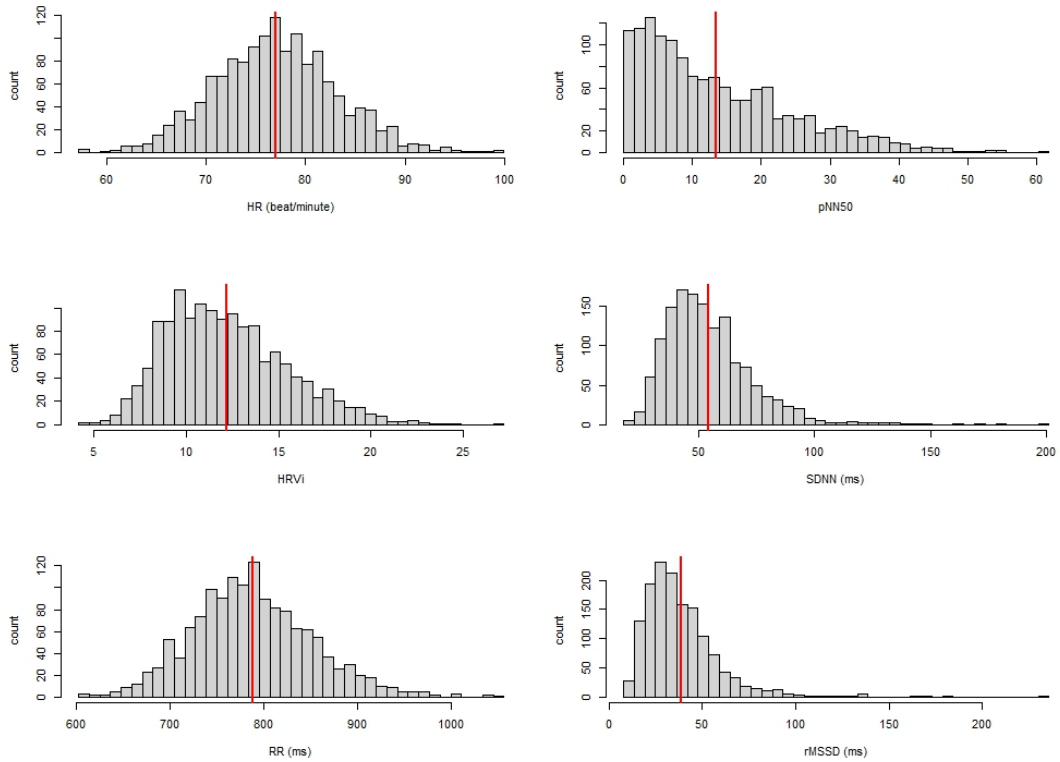


Figure 3: Distributions of the time-domain HRV indices over all the daily five-minute recordings. The mean of each distribution is indicated by a red line.

2.2.4 Blood pressure and lean test

2.2.5 Body composition

2.2.6 Indoor and outdoor environment

2.2.7 EEG and fNIRS

2.2.8 Blood tests

2.2.9 Drugs and supplements

2.3 Data processing

2.4 Statistical analysis

3 Results

Table 1: CAPTION.

Variable	Value	Unit
NAME	0.0	ms

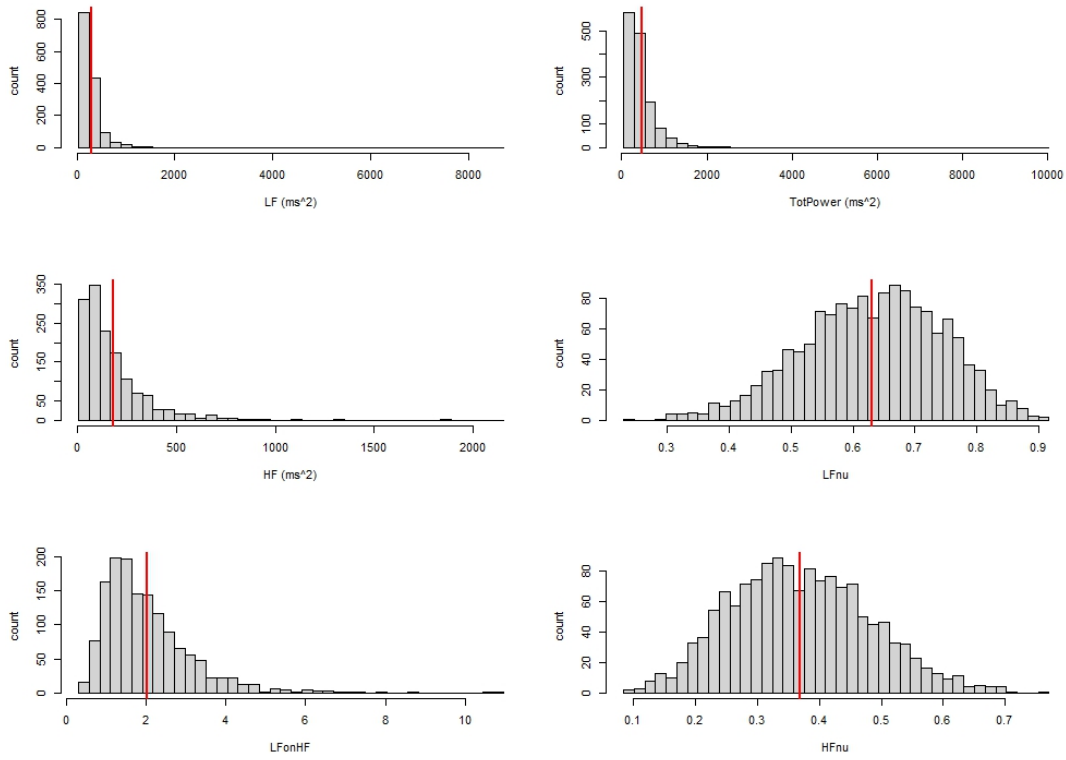


Figure 4: Distributions of the frequency-domain HRV indices over all the daily five-minute recordings. LFonHF is the ratio LF/HF; TotPower is LF + HF. The mean of each distribution is indicated by a red line.

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Figure 5: CAPTION.

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(a) CAPTION A.

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(b) CAPTION B.

Figure 6: CAPTION OF THE PAIR.

4 Discussion

4.1 Limitations

5 Conclusions

Data and code availability

Code: `https://github.com/USERNAME/N1-LoSt`.

Acknowledgements

References

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A Supplementary material